

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech (CE) Examination****May 2018****CE307.02 Advanced Java Programming****Date: 10.05.2018, Thursday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q - 1 Answer the following questions.**

- A.** Justify “UDP is protocol that transfers data in a connection less and unreliable manner on a network.” **[01]**
- B.** Differentiate between GenericServlet and HttpServlet **[01]**
- C.** Compare JSP and Servlet. **[02]**
- D.** Define Servlet. Draw Servlet life Cycle. **[03]**
- E.** What is deployment descriptor? Give its importance and provide a sample web.xml file for simple servlet. **[04]**

Q – 2.A List the different types of JDBC driver and explain Type 4 with its advantages and disadvantages. **[04]**

Q – 2.B Answer the following questions [Any Two]. **[08]**

- A.** Explain following JDBC ResultSet Methods with example.
1. absolute() 2. relative()
- B.** What is RequestDispatcher object? Explain forward() and include() methods and compare them.
- C.** What is filter? Explain Filer, FilteChain and FilterConfig interfaces.

Q - 3 Answer the following questions. [Any Two]. **[12]**

- A.** Write a client server program using TCP where client sends 10 numbers and server responds with number in sorted order.
- B.** Write a JDBC program to deletes a student record whose roll number is given as input.
- C.** Write a servlet to store person details(name, email-Id) as cookie and then display it by reading from cookie.[**Note:** set cookie age for 1 year]

SECTION – II**Q - 4 Answer the following questions.**

A. What is the output of the following code? **[02]**

```
<html><body>
<% int i = 10 ;%>
<% while(--i>=0) { %>
    out.print(i);
<% } %> </body></html>
```

B. What will be the output of the following code? **[02]**

```
<html><body>
<% x=3; %>
<% int x=5; %>
<%! int x=7; %>
x = <%=x%>, <%=this.x%> </body></html>
```

C. What is JSTL? Which method of the simple tag handler is used to process the tag? **[02]**

D. List and explain scripting elements supported in Java Server Pages. **[02]**

E. List JSP implicit objects. Explain application and config with example. **[03]**

Q – 5.A What is Facelets? Explain the life cycle of Facelets. List its advantages. **[06]**

Q – 5.B Write a JSF code for following task. **[06]**

1. Create a required input-text with auto-complete off and display required message.
2. Display output in red color, 12px font-size.
3. Convert input number into (###, ###, ###) pattern.

OR

Q – 5.A Write a JavaBean named "Student" having roll-number and name having getter and setter methods. Write a JSP page to set the roll-number and name for a student object and then print them by reading from object. **[06]**

Q – 5.B Create a login page (username, password) and signup page(username, password, confirm-password) using JSP store/verify necessary field to/from database. **[06]**

Q – 6. Answer the following. [Attempt any Two] **[12]**

- A.** Define Hibernate. Draw and explain Hibernate cache architecture.
- B.** What is HQL? How does it differ from SQL? List its advantages.
- C.** Define java Spring. Explain Aspect oriented programming (AOP) framework.
- D.** What is Dependency Injection (DI)? Explain Constructor Injection(CI) with example.